

Rigidity at the Extremal Crouzeix Constant

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1 Abstract

For a complex matrix A , let $\psi(A)$ be the norm of the polynomial functional calculus on its numerical range. We prove that $\psi(A) = 2$ only when the numerical range is a nondegenerate closed disk. The proof first extracts the equality conditions in the sharp double-layer estimate. After boundary eigenvalues are removed, an extremizer obtained from a finite Blaschke product satisfies an exact boundary-kernel identity. On a curved exposed arc this identity makes the extremizer rational. A direct resolvent argument then places every zero of its reduced numerator inside the numerical range and every pole outside; consequently its full unit lemniscate coincides with the boundary of the numerical range. The defining polynomial of the projective dual of that algebraic boundary divides the Hermitian Kippenhahn determinant. Hyperbolicity and strict convexity force the dual curve to have degree two, and its points at infinity force the resulting conic to be a circle.

Keywords: numerical range; Crouzeix constant; spectral set; rational lemniscate; hyperbolic polynomial

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1. Introduction

Throughout, $\mathbb{C}$ and $\mathbb{R}$ denote the complex and real fields, $M_N(\mathbb{C})$ the algebra of $N \times N$ complex matrices, and I the identity matrix of the size dictated by context. For a matrix B , the symbols B^* , $\text{spec}(B)$, and $\text{spr}(B)$ denote its conjugate transpose, spectrum, and spectral radius. The notation $\|\cdot\|$ denotes the Hilbert-space norm on vectors and the induced operator norm on bounded linear maps; on $\mathbb{C}^N$ these are the Euclidean and induced Euclidean norms. We put

$$\text{Re } B = \frac{B + B^*}{2}, \quad \text{Im } B = \frac{B - B^*}{2i}.$$

For scalars and functions, Re and Im have their usual pointwise meanings. For Hermitian matrices B, C , the relation $B \succeq C$ means that $B - C$ is positive semidefinite. We write $\text{int } E$ for the interior of a planar set E and $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ for the open unit disk. Scalar matrix coefficients are written as $x^*By = \langle By, x \rangle$, where inner products are linear in the first argument. For a bounded scalar-valued function g on a set E , write $\|g\|_E = \sup_{z \in E} |g(z)|$.

For $A \in M_N(\mathbb{C})$, set

$$W(A) = \{u^*Au : u \in \mathbb{C}^N, \|u\| = 1\}.$$

For a polynomial p , put

$$\psi(A) = \sup \left\{ \|p(A)\| : p \in \mathbb{C}[z], \|p\|_{W(A)} \leq 1 \right\}.$$

For a compact convex body K , let $C(K)$ denote the complex-valued continuous functions on K , and write

$$\mathcal{A}(K) = \{f \in C(K) : f|_{\text{int } K} \text{ is holomorphic}\}.$$

The space $H^\infty(\text{int } K)$ consists of the bounded holomorphic functions on $\text{int } K$ and carries the norm $\|\cdot\|_{\text{int } K}$. Whenever $\text{spec}(A) \subset \text{int } K$, the notation $f(A)$ denotes the standard finite-dimensional holomorphic functional calculus, equivalently evaluation of any Hermite interpolating polynomial with the required spectral jets. The constant $\psi(A)$ was introduced by Crouzeix [1], who conjectured that his dimension-two disk-extremality result persists in every dimension, which would imply $\psi(A) \leq 2$. The operator-valued integral representation behind the modern estimates goes back to Delyon and Delyon [2]. Crouzeix and Palencia proved the bound $1 + \sqrt{2}$ [3], and Lorist and Schwenninger established the conjectured bound 2 [4]. We retain the equality information in the double-layer and iterate arguments. The resulting rigidity statement is the following.

Theorem 1. *For every $N \geq 1$ and $A \in M_N(\mathbb{C})$,*

$$\psi(A) = 2 \implies W(A) = \{z \in \mathbb{C} : |z - \gamma| \leq \rho\}$$

for some $\gamma \in \mathbb{C}$ and some $\rho > 0$.

The proof has four steps. An equality analysis of the sharp bound produces two extremal vectors and a pointwise boundary-kernel identity. A curved exposed arc then turns that identity into a rational formula for the extremizer. A direct resolvent argument identifies its full unit lemniscate with $\partial W(A)$. Finally, the Kippenhahn determinant contains the dual curve as a hyperbolic factor. A generic contact count makes the dual a conic, and the circular points at infinity then force $\partial W(A)$ to be a circle.

2. Numerical-range preliminaries

Here and below, $\text{conv } E$ denotes the convex hull of E . For a Hermitian matrix B , $\lambda_{\max}(B)$ denotes its largest eigenvalue. At a differentiability point σ of a convex boundary, $\mathbf{n}(\sigma)$ denotes its outward unit normal; ν , with or without a subscript, denotes an arbitrary supporting unit normal.

Lemma 1 (Numerical-range geometry). *The following facts hold.*

1. $W(A)$ is nonempty, compact, and convex.
2. If $W(A)$ has empty interior, then A is normal and $\psi(A) = 1$.
3. For $\alpha \in \mathbb{C} \setminus \{0\}$, $\beta \in \mathbb{C}$, and unitary $\mathcal{U}$,

$$W(\alpha A + \beta I) = \alpha W(A) + \beta, \quad \psi(\alpha A + \beta I) = \psi(A), \quad \psi(\mathcal{U}^* A \mathcal{U}) = \psi(A).$$

Also $W(A^*) = \overline{W(A)}$ and $\psi(A^*) = \psi(A)$.

4. For a finite nonempty direct sum,

$$W\left(\bigoplus_j A_j\right) = \text{conv}\left(\bigcup_j W(A_j)\right), \quad \psi\left(\bigoplus_j A_j\right) \leq \max_j \psi(A_j).$$

5. The support function of $W(A)$ is

$$h_A(\theta) = \lambda_{\max}\left(\text{Re}(e^{-i\theta} A)\right).$$

Proof. The unit sphere is compact and $u \mapsto u^* A u$ is continuous, proving nonemptiness and compactness. To prove convexity, take two points of $W(A)$ and compress A to the span of corresponding unit vectors. If the points are distinct, those vectors are linearly independent; if they coincide, there is no segment to prove. It is therefore enough to know that the numerical range of a 2×2 matrix is convex.

After unitary triangularization such a matrix has the form $\begin{pmatrix} \lambda_1 & \tau \\ 0 & \lambda_2 \end{pmatrix}$. For $u = (\cos t, e^{i\varphi} \sin t)$, where $0 \leq t \leq \pi/2$ and $\varphi \in \mathbb{R}$,

$$u^* A u = \lambda_1 \cos^2 t + \lambda_2 \sin^2 t + \tau e^{i\varphi} \sin t \cos t.$$

These points fill the possibly degenerate ellipse with foci λ_1, λ_2 , hence a convex set. The two-dimensional compression is contained in $W(A)$, so the segment joining the original two points is contained in $W(A)$.

If $W(A)$ has empty interior, its convexity places it in an affine line. If $W(A) = \{\gamma\}$, then $x^*(A - \gamma I)x = 0$ for all x . The complex polarization identity makes every matrix coefficient of $A - \gamma I$ zero. If $W(A)$ lies in a line, a translation and a nonzero complex rotation make all quadratic forms of A real. The skew-Hermitian part then has zero quadratic form and vanishes by polarization, so the transformed matrix is Hermitian. A normal matrix satisfies

$$\|p(A)\| = \|p\|_{\text{spec}(A)} \leq \|p\|_{W(A)}.$$

The constant polynomial $p \equiv 1$ gives the reverse inequality.

The affine and unitary formulas follow by changing variables in polynomials. For adjoints use $\tilde{p}(z) = \overline{p(\bar{z})}$, for which $\tilde{p}(A^*) = p(A)^*$. For a direct sum, a unit vector decomposes into block components and its numerical value is a convex combination of block numerical values. Also $p(\oplus_j A_j) = \oplus_j p(A_j)$; the supremum norm on the convex hull dominates that on every block. Finally,

$$\max_{z \in W(A)} \text{Re}(e^{-i\theta} z) = \max_{\|u\|=1} u^* \text{Re}(e^{-i\theta} A) u,$$

which is the largest eigenvalue of the displayed Hermitian matrix. $\square$

3. The sharp bound and its equality case

The equality analysis begins with the iterated-dilation estimate of Lorist and Schwenninger [4]; we first extract the equality consequences needed below.

Lemma 2 (An abstract dilation estimate). *Let T act on a nonzero finite-dimensional Hilbert space. Suppose that $\mathcal{M}$ is a contraction on another Hilbert space, $\mathcal{V}$ is an isometry, and*

$$E_n = 2\mathcal{V}^* \mathcal{M}^{*n} \mathcal{V} - T^{*n} \quad (n \geq 1)$$

are uniformly bounded and commute with T . Then $\|T\| \leq 2$.

If additionally $\|T\| = 2$, $\text{spr}(T) < 1$, $T^*Tx = 4x$, $\|x\| = 1$, and $y = Tx/2$, then

$$T^*x = 0, \quad T^*y = 2x, \quad \mathcal{M}\mathcal{V}x = \mathcal{V}y, \quad \mathcal{M}^*\mathcal{V}y = \mathcal{V}x.$$

Proof. Put $\kappa = \|T\|$. If $\kappa \leq 1$, the conclusion is immediate; assume $\kappa > 1$. Choose a unit vector x with $T^*Tx = \kappa^2x$, and set

$$m_n = \text{Re}(x^* E_n T^n x), \quad \omega = \mathcal{M}^* \mathcal{V} T x - \kappa \mathcal{V} x.$$

The defining identity gives $\|T^{*n}\| \leq 2 + \sup_k \|E_k\|$, so the powers of T , and hence (m_n) , are bounded.

Put $Q_n = \mathcal{V}^* \mathcal{M}^{*n} \mathcal{V}$ and $q_n = T^{*n}x$. Commutativity of E_n with T gives

$$2(Q_{n+1}T - TQ_{n+1}) = T^{*(n+1)}T - TT^{*(n+1)}.$$

Moreover, $m_n = 2 \operatorname{Re}(q_n^* Q_n x) - \|q_n\|^2$ and $\mathcal{V}^* \mathcal{M}^{*n} \omega = Q_{n+1} T x - \kappa Q_n x$. Multiplying the displayed commutator identity on the left by q_n^* , using $T^* T x = \kappa^2 x$, and taking real parts now gives

$$\begin{aligned} \kappa m_n - m_{n+1} &= \kappa(\kappa - 1) \|T^{*n} x\|^2 - 2 \operatorname{Re}((\mathcal{V}^* \mathcal{M}^{*n} \omega)^* T^{*n} x) \\ &= \kappa(\kappa - 1) \left\| T^{*n} x - \frac{\mathcal{V}^* \mathcal{M}^{*n} \omega}{\kappa(\kappa - 1)} \right\|^2 - \frac{\|\mathcal{V}^* \mathcal{M}^{*n} \omega\|^2}{\kappa(\kappa - 1)} \\ &\geq -\frac{\|\omega\|^2}{\kappa(\kappa - 1)}. \end{aligned}$$

103 Telescoping, with the boundedness of m_n , yields

$$104 \quad \kappa m_1 = \sum_{n \geq 1} \kappa^{-n+1} (\kappa m_n - m_{n+1}) \geq -\frac{\|\omega\|^2}{(\kappa - 1)^2}. \quad (3.1)$$

On the other hand, using that $\mathcal{V}$ is isometric and $\mathcal{M}$ contractive,

$$\begin{aligned} \|\omega\|^2 &\leq 2\kappa^2 - 2\kappa \operatorname{Re}(x^* \mathcal{V}^* \mathcal{M}^* \mathcal{V} T x) \\ &= 2\kappa^2 - \kappa m_1 - \kappa^3. \end{aligned} \quad (3.2)$$

105 Combining (3.1) and (3.2) gives

$$106 \quad \|\omega\|^2 \left(1 - \frac{1}{(\kappa - 1)^2} \right) \leq \kappa^2 (2 - \kappa),$$

107 which is impossible for $\kappa > 2$.

108 Now let $\kappa = 2$. Equations (3.1)–(3.2) become $2m_1 \geq -\|\omega\|^2$ and $\|\omega\|^2 \leq -2m_1$, so equality holds
109 in both. For each n , the slack in the estimate preceding (3.1) is

$$110 \quad 2 \left\| T^{*n} x - \frac{1}{2} \mathcal{V}^* \mathcal{M}^{*n} \omega \right\|^2 + \frac{1}{2} (\|\omega\|^2 - \|\mathcal{V}^* \mathcal{M}^{*n} \omega\|^2).$$

111 Their positively weighted sum is zero. Thus, for every n ,

$$112 \quad \mathcal{V}^* \mathcal{M}^{*n} \omega = 2T^{*n} x, \quad \|\mathcal{V}^* \mathcal{M}^{*n} \omega\| = \|\omega\|.$$

113 Since $\operatorname{spr}(T) < 1$, $T^{*n} x \rightarrow 0$, and these two equalities force $\omega = 0$. Their first instance then gives
114 $T^* x = 0$. With $y = Tx/2$, $\mathcal{M}^* \mathcal{V} y = \mathcal{V} x$. Both $\mathcal{V} x$ and $\mathcal{V} y$ are unit vectors, so equality in Cauchy–
115 Schwarz for $\mathcal{M}$ gives $\mathcal{M} \mathcal{V} x = \mathcal{V} y$. Finally $T^* y = T^* T x / 2 = 2x$. $\square$

116 **Proposition 1** (The sharp numerical-range estimate). *For every $N \geq 1$, every $A \in M_N(\mathbb{C})$, and every
117 polynomial p ,*

$$118 \quad \|p(A)\| \leq 2\|p\|_{W(A)}. \quad (3.3)$$

119 *More generally, if K is a compact convex body, $W(A) \subset K$, $\operatorname{spec}(A) \subset \operatorname{int} K$, and $f \in \mathcal{A}(K)$, then*

$$120 \quad \|f(A)\| \leq 2\|f\|_K. \quad (3.4)$$

121 *For $f \in H^\infty(\operatorname{int} K)$, the corresponding estimate is*

$$122 \quad \|f(A)\| \leq 2\|f\|_{\operatorname{int} K}.$$

123 **Proof.** Normalize $\|f\|_K \leq 1$. Cauchy's boundary formula below also holds when f is only in $\mathcal{A}(K)$:
124 for $z_0 \in \operatorname{int} K$, the functions $f(z_0 + r(z - z_0))$, $r < 1$, are holomorphic on a neighborhood of K
125 and converge uniformly to f ; apply the usual formula to them and pass to the limit. The positive
126 operator-valued double-layer kernel used below originates in Delyon and Delyon [2] and appears in

the Cauchy-transform form of Crouzeix and Palencia [3]. Parametrize the rectifiable convex Jordan curve $\Gamma = \partial K$ counterclockwise by arclength. Its outward unit normal $\mathbf{n}(\sigma)$ exists almost everywhere. Put $R_\sigma = (\sigma I - A)^{-1}$ and

$$\mathcal{P}(\sigma) = \frac{1}{\pi} \operatorname{Re}(\mathbf{n}(\sigma) R_\sigma).$$

At a point with a normal, the supporting-half-plane property gives

$$\Delta_\sigma := \operatorname{Re}(\overline{\mathbf{n}(\sigma)}(\sigma I - A)) \succeq 0.$$

Multiplication by the two resolvents shows

$$\operatorname{Re}(\mathbf{n}(\sigma) R_\sigma) = R_\sigma^* \Delta_\sigma R_\sigma \succeq 0. \quad (3.5)$$

Since $d\sigma = i\mathbf{n}(\sigma) ds$ and $d\bar{\sigma} = -i\overline{\mathbf{n}(\sigma)} ds$,

$$\mathcal{P}(\sigma) ds = \frac{1}{2\pi i} R_\sigma d\sigma - \frac{1}{2\pi i} R_\sigma^* d\bar{\sigma}.$$

Cauchy's formula gives $\int_\Gamma R_\sigma d\sigma = 2\pi i I$; taking adjoints gives $\int_\Gamma R_\sigma^* d\bar{\sigma} = -2\pi i I$. Therefore

$$\int_\Gamma \mathcal{P}(\sigma) ds = 2I. \quad (3.6)$$

Consequently

$$(\mathcal{V}x)(\sigma) = 2^{-1/2} \mathcal{P}(\sigma)^{1/2} x$$

defines an isometry into $L^2(\Gamma; \mathbb{C}^N)$. Let M_f be multiplication by f , and set $T = f(A)$. For every $n \geq 1$, Cauchy's formula and the preceding decomposition give

$$2\mathcal{V}^* M_f^{*n} \mathcal{V} = \frac{1}{2\pi i} \int_\Gamma \overline{f(\sigma)^n} R_\sigma d\sigma + T^{*n},$$

because

$$\int_\Gamma \overline{f(\sigma)^n} R_\sigma^* d\bar{\sigma} = \left(\int_\Gamma f(\sigma)^n R_\sigma d\sigma \right)^* = -2\pi i T^{*n}.$$

Thus

$$E_n := 2\mathcal{V}^* M_f^{*n} \mathcal{V} - T^{*n} = \frac{1}{2\pi i} \int_\Gamma \overline{f(\sigma)^n} R_\sigma d\sigma. \quad (3.7)$$

The right side commutes with T , because every resolvent does. Also

$$\sup_n \|E_n\| \leq \frac{\operatorname{length}(\Gamma)}{2\pi} \max_{\sigma \in \Gamma} \|R_\sigma\| < \infty.$$

Lemma 2 proves (3.4).

For $f \in H^\infty(\operatorname{int} K)$, normalize $\|f\|_{\operatorname{int} K} \leq 1$, choose a Riemann map $\phi : \operatorname{int} K \rightarrow \mathbb{D}$, and put $g = f \circ \phi^{-1}$. For $0 < r < 1$, define

$$f_r(z) = g(r\phi(z)).$$

These functions belong to $\mathcal{A}(K)$, have norm at most 1, and their spectral jets converge to those of f , because $g(r \cdot) \rightarrow g$ locally uniformly with all derivatives. Hence $f_r(A) \rightarrow f(A)$, proving the H^∞ estimate.

Finally, for an arbitrary A , apply (3.4) to the outer parallel convex body $K_\varepsilon = W(A) + \varepsilon \overline{\mathbb{D}}$, which contains $W(A)$ in its interior. For a polynomial p , $\|p\|_{K_\varepsilon} \rightarrow \|p\|_{W(A)}$. Letting $\varepsilon \downarrow 0$ proves (3.3), including all boundary-spectrum and nonsmooth cases. $\square$

Proposition 2 (Equality data). *Let $K = W(A)$ have nonempty interior, let $\text{spec}(A) \subset \Omega := \text{int } K$, and let $f \in \mathcal{A}(K)$ be nonconstant with*

$$\|f\|_K = 1, \quad |f| = 1 \text{ on } \partial K, \quad \|f(A)\| = 2.$$

Then there are orthonormal vectors x, y such that, with $T = f(A)$,

$$Tx = 2y, \quad T^*x = 0, \quad T^*y = 2x, \quad (3.8)$$

and, for almost every $\sigma \in \partial K$,

$$\mathcal{P}(\sigma)^{1/2}(y - f(\sigma)x) = 0. \quad (3.9)$$

Proof. Apply the construction in Proposition 1. The spectral mapping theorem gives $\text{spr}(T) < 1$, because the finitely many eigenvalues of A are inside Ω and a nonconstant inner function has modulus strictly less than 1 there. Choose x with $T^*Tx = 4x$, and put $y = Tx/2$. Lemma 2 gives $T^*x = 0$, $T^*y = 2x$, and $M_f \mathcal{V}x = \mathcal{V}y$. Here $\|y\| = 1$, and $2x^*y = x^*Tx = (T^*x)^*x = 0$, so x, y are orthonormal. The multiplication identity is exactly (3.9). $\square$

Lemma 3 (Boundary eigenvalues reduce). *If $Ax = \lambda x$, $\|x\| = 1$, and $\lambda \in \partial W(A)$, then $A^*x = \bar{\lambda}x$. Consequently, with $\simeq$ denoting unitary similarity,*

$$A \simeq A_\partial \oplus A_{\text{in}},$$

where A_∂ is diagonal, with the boundary eigenvalues as its diagonal entries, while $\text{spec}(A_{\text{in}}) \subset \text{int } W(A)$.

Proof. Choose a supporting direction ν , $|\nu| = 1$, at λ . Equality in the Rayleigh quotient gives

$$\text{Re}(\bar{\nu}A)x = \text{Re}(\bar{\nu}\lambda)x.$$

Substituting $Ax = \lambda x$ gives $A^*x = \bar{\lambda}x$. Thus $\mathbb{C}x$ reduces A . Split it off and repeat. Every eigenvalue belongs to $W(A)$, since an eigenvector realizes it, so the remaining spectrum is in the interior. $\square$

Lemma 4 (Conformal and finite-interpolation facts). *Let Ω be a bounded convex domain.*

1. *There is a conformal bijection $\phi : \Omega \rightarrow \mathbb{D}$, and it extends to a homeomorphism $\bar{\Omega} \rightarrow \bar{\mathbb{D}}$.*
2. *At each of finitely many distinct points of Ω , prescribe a contiguous jet, consisting of the value and all derivatives through a fixed finite order. If these data are realized by a function of $H^\infty(\Omega)$ of norm at most 1, they are also realized by $\mathcal{B} \circ \phi$, where $\mathcal{B}$ is a finite Blaschke product. If the infimum of the supremum norms over all interpolants of the prescribed data is 1, then the Schur-class interpolant is unique and is of this form.*

Proof. A bounded convex domain is a Jordan domain. The Riemann mapping theorem and the Carathéodory boundary theorem therefore give the first assertion.

Transfer the jet data to $\mathbb{D}$ by ϕ^{-1} . The confluent Nevanlinna–Pick theorem says that such finite data are feasible in the Schur class exactly when their generalized Pick matrix is positive semidefinite. The Schur recursion, with the final free parameter chosen unimodular, then produces a rational inner solution, hence a finite Blaschke product. More precisely, let $\mathcal{J}$ be the finite linear map that sends a holomorphic function to the prescribed vector $\mathbf{d}$ of values and derivatives, and put

$$c_0 = \inf\{\|g\|_\Omega : g \in H^\infty(\Omega), \mathcal{J}g = \mathbf{d}\}.$$

Before forming the generalized Pick matrix, discard any duplicate conditions. The remaining evaluation and derivative functionals are linearly independent. If $c_0 = 1$, the resulting Pick matrix at norm one is singular: if it were positive definite, continuity would preserve positive semidefiniteness when the

norm bound 1 is decreased slightly, contrary to the definition of c_0 . The uniqueness theorem for a solvable singular confluent Pick problem now applies, and the Schur recursion shows that its unique Schur solution is a finite Blaschke product. The matrix-induced version of these facts, including the repeated-node conditions arising from Jordan blocks, is due to Crouzeix [1]. $\square$

For a bounded domain Ω and a matrix A_0 with $\text{spec}(A_0) \subset \Omega$, define

$$\psi_\Omega(A_0) = \sup\{\|g(A_0)\| : g \in H^\infty(\Omega), \|g\|_\Omega \leq 1\}.$$

The value $g(A_0)$ depends only on finitely many derivatives, those prescribed by the Jordan blocks. Montel's theorem therefore makes the supremum a maximum. When Ω is convex, Lemma 4 replaces an extremizer by a finite Blaschke product composed with a Riemann map, without changing $g(A_0)$.

Lemma 5 (Strict domain monotonicity). *Let $\Omega_0 \subsetneq \Omega_1$ be bounded convex domains and suppose $\text{spec}(A_0) \subset \Omega_0$. If $\psi_{\Omega_1}(A_0) > 1$, then*

$$\psi_{\Omega_0}(A_0) > \psi_{\Omega_1}(A_0).$$

Proof. Monotonicity in the displayed direction is immediate by restriction. Suppose equality held, with common value $\eta > 1$, and choose a norm-one extremizer g on Ω_1 . Its norm on Ω_0 must be exactly 1, since otherwise rescaling its restriction gives a value larger than η . Thus it is also an extremizer on Ω_0 .

For either domain, interpolate the finite jet of g that determines $g(A_0)$ with least possible norm. A norm smaller than 1 would contradict extremality after rescaling. The uniqueness part of Lemma 4 identifies g on each domain with a finite Blaschke product composed with its Riemann map. In particular, the restriction of g to Ω_0 has a continuous extension to $\overline{\Omega_0}$ with modulus one on $\partial\Omega_0$.

Choose $z_* \in \Omega_0$. Along every ray from z_* , convexity gives an exit radius for each domain. Since the domains differ, on some ray the first exit occurs strictly earlier; hence there is $\xi \in \partial\Omega_0 \cap \Omega_1$. At this interior point of Ω_1 , continuity of the original holomorphic function and of the boundary extension gives $|g(\xi)| = 1$. The maximum-modulus principle makes g constant, which would give $\eta = 1$, a contradiction. $\square$

4. Reduction to an interior-spectrum equality pair

Lemma 6 (Boundary-spectrum induction). *Assume Theorem 1 has been proved in every size smaller than N . Let $A \in M_N(\mathbb{C})$, let $K = W(A)$, and suppose $\psi(A) = 2$. Then either K is a nondegenerate closed disk, or*

$$\text{spec}(A) \subset \text{int } K. \tag{4.1}$$

Proof. By Lemma 1, K has nonempty interior. Choose polynomials p_j such that

$$\|p_j\|_K = 1, \quad \|p_j(A)\| \longrightarrow 2. \tag{4.2}$$

Suppose A has a boundary eigenvalue. Lemma 3 gives $A \simeq A_\partial \oplus A_1$, where A_∂ is diagonal with spectrum on ∂K and $\text{spec}(A_1) \subset \text{int } K$. The A_∂ -block of $p_j(A)$ has norm at most 1, so

$$\|p_j(A_1)\| \longrightarrow 2. \tag{4.3}$$

In particular A_1 is nonempty. Since $W(A_1) \subset K$, (4.3) and Proposition 1 give $\psi(A_1) = 2$. Its size is smaller than N , so induction says that

$$K_1 := W(A_1)$$

is a closed disk of positive radius.

Apply Lemma 3 once more, now to A_1 relative to K_1 :

$$A_1 \simeq A_{\partial,1} \oplus A_2,$$

where $A_{\partial,1}$ is a finite diagonal block on ∂K_1 and $\text{spec}(A_2) \subset \text{int } K_1$. The same sequence (4.2) is bounded by 1 at the eigenvalues of $A_{\partial,1}$, all of which lie in K , so $\|p_j(A_2)\| \rightarrow 2$. Thus A_2 is nonempty.

Write $\Lambda_1 = \text{spec}(A_{\partial,1})$. Since

$$K_1 = W(A_1) = \text{conv}(\Lambda_1 \cup W(A_2)),$$

every extreme point of K_1 belongs to the compact set $\Lambda_1 \cup W(A_2)$. Indeed, a finite convex representation of an extreme point can contain no two distinct points with positive coefficients. The circle ∂K_1 has infinitely many extreme points and Λ_1 is finite. Hence $\partial K_1 \setminus \Lambda_1 \subset W(A_2)$; closedness and convexity of $W(A_2)$ give

$$W(A_2) = K_1. \quad (4.4)$$

Let $\Omega_0 = \text{int } K_1$ and $\Omega_1 = \text{int } K$. The polynomial sequence and Proposition 1 show

$$\psi_{\Omega_0}(A_2) = \psi_{\Omega_1}(A_2) = 2. \quad (4.5)$$

Indeed,

$$\|p_j\|_{\Omega_0} = \|p_j\|_{K_1} \leq 1, \quad \|p_j\|_{\Omega_1} = \|p_j\|_K = 1,$$

so the lower bounds follow from (4.2)–(4.4), while the upper bounds follow from the H^∞ estimate in Proposition 1. If $K_1 \subsetneq K$, then $\Omega_0 \subsetneq \Omega_1$, and Lemma 5 contradicts (4.5). Therefore $K_1 = K$, and K is the asserted disk. If this conclusion does not occur, A has no boundary eigenvalue, which is (4.1). $\square$

Lemma 7 (The holomorphic extremizer). *Suppose $K = W(A)$ has nonempty interior and $\text{spec}(A) \subset \Omega = \text{int } K$. If $\psi(A) = 2$, there is a nonconstant*

$$f = \mathcal{B} \circ \phi \in \mathcal{A}(K),$$

where $\phi : \Omega \rightarrow \mathbb{D}$ is conformal and $\mathcal{B}$ is a finite Blaschke product, such that

$$\|f\|_K = 1, \quad |f| = 1 \text{ on } \partial K, \quad \|f(A)\| = 2. \quad (4.6)$$

Proof. Normalized polynomials witnessing $\psi(A) = 2$ show $\psi_\Omega(A) \geq 2$, and Proposition 1 gives the reverse inequality. Montel compactness, followed by convergence of the finitely many spectral jets, gives an $H^\infty(\Omega)$ maximizer of norm at most one. Its norm is exactly one, since otherwise rescaling it would produce a matrix value of norm greater than two. Apply the finite Schur algorithm in Lemma 4 to the jet that determines its matrix value. It supplies $\mathcal{B} \circ \phi$ with the same matrix value. The boundary extension of ϕ puts this function in $\mathcal{A}(K)$ and gives boundary modulus 1. It cannot be constant because a constant matrix has norm at most 1. $\square$

5. From equality to a rational inner function

We first locate an analytic boundary arc.

Lemma 8 (A curved numerical-range arc). *If $K = W(A)$ has interior, $\text{spec}(A) \subset \text{int } K$, and K is not a polygon, then ∂K contains an open real-analytic arc on which every supporting line exposes one point and the curvature is positive. Moreover, under $\text{spec}(A) \subset \text{int } K$, K cannot be a polygon.*

Proof. Put

$$\mathcal{H}(\theta) = \text{Re}(e^{-i\theta} A), \quad h_A(\theta) = \lambda_{\max}(\mathcal{H}(\theta)).$$

By Rellich's analytic perturbation theorem [5], for every θ_0 the eigenvalues of the real-analytic Hermitian matrix family $\mathcal{H}(\theta)$ can be numbered, with multiplicity, by real-analytic functions of θ on a neighborhood of θ_0 . Take a finite cover of the parameter circle by such intervals and merge branches that agree identically. On the closure of a smaller interval, the difference of two remaining branches has only finitely many zeros by the analytic identity theorem. Subdividing at these crossings gives a finite partition into intervals on which h_A is analytic.

At a differentiability point, the exposed boundary point is

$$\sigma(\theta) = e^{i\theta}(h_A(\theta) + ih'_A(\theta)), \quad \sigma'(\theta) = ie^{i\theta}(h_A(\theta) + h''_A(\theta)). \quad (5.1)$$

The first identity follows by differentiating the top Rayleigh quotient; the second follows by differentiation. Convexity gives $h_A + h''_A \geq 0$. If this function vanished identically on every analytic interval, σ would be constant on each interval and K would be a polygon. Otherwise analyticity supplies a subinterval on which $h_A + h''_A > 0$. Formula (5.1) then gives the required regular, positively curved arc. Differentiability of the support function makes the exposed face a singleton there.

It remains to exclude a polygon when the spectrum is interior. Let λ be a vertex and let a unit vector v realize $v^*Av = \lambda$. Two linearly independent supporting normals v_1, v_2 at the vertex give

$$\operatorname{Re}(\bar{v}_j A)v = \operatorname{Re}(\bar{v}_j \lambda)v \quad (j = 1, 2).$$

The two independent real-linear equations imply $Av = \lambda v$ and $A^*v = \bar{\lambda}v$. Thus λ is a boundary eigenvalue, contradicting $\operatorname{spec}(A) \subset \operatorname{int} K$. $\square$

Proposition 3 (Rational collapse). *Under the hypotheses and notation of Lemma 7, let x, y be the equality vectors from Proposition 2. Then the function f is rational on Ω . More precisely, for $z \notin \operatorname{spec}(A)$, set*

$$a(z) = x^*(zI - A)^{-1}x, \quad c(z) = y^*(zI - A)^{-1}x,$$

and regard a, c thereafter as their meromorphic adjugate continuations. Then

$$c(z)f(z) = 2a(z) \quad (5.2)$$

as an identity of meromorphic functions on Ω . Equivalently, $f = 2a/c$ wherever the quotient is defined, and then everywhere after removable cancellation. After cancellation, $f = U/V$ for coprime polynomials with

$$\deg U > \deg V. \quad (5.3)$$

If adj denotes the classical adjugate and

$$D_A(z) = x^* \operatorname{adj}(zI - A)x, \quad C_A(z) = y^* \operatorname{adj}(zI - A)x,$$

then the reduction can be chosen so that

$$U(z)C_A(z) = 2D_A(z)V(z).$$

Proof. Choose the arc from Lemma 8; it avoids the finite spectrum. For almost every point σ of that arc, let n be its outward normal, $R = (\sigma I - A)^{-1}$, and

$$\Delta_\sigma = \operatorname{Re}(\bar{n}(\sigma I - A)) \succeq 0.$$

Equations (3.5) and (3.9) imply

$$\Delta_\sigma R(y - f(\sigma)x) = 0. \quad (5.4)$$

Indeed, squaring the norm in (3.9) and using $\mathcal{P}(\sigma) = \pi^{-1}R^*\Delta_\sigma R$ first gives $\Delta_\sigma^{1/2}R(y - f(\sigma)x) = 0$, and hence (5.4). The vector $r_\sigma = R(y - f(\sigma)x)$ is nonzero: R is invertible, while the orthonormality of x, y makes $y - f(\sigma)x \neq 0$. It is therefore an eigenvector of $\operatorname{Re}(\bar{n}A)$ for its maximal eigenvalue. The exposed point is unique, so $r_\sigma^*Ar_\sigma = \sigma\|r_\sigma\|^2$. Since $(\sigma I - A)r_\sigma = y - f(\sigma)x$, taking the inner product with r_σ and conjugating gives

$$(y - f(\sigma)x)^*R(y - f(\sigma)x) = 0. \quad (5.5)$$

For $z \notin \operatorname{spec}(A)$, put $R_z = (zI - A)^{-1}$. This resolvent commutes with $T = f(A)$, and relations (3.8) give

$$x^*R_z y = \frac{1}{2}x^*R_z T x = \frac{1}{2}x^*T R_z x = 0, \quad y^*R_z y = \frac{1}{2}y^*R_z T x = \frac{1}{2}y^*T R_z x = a(z). \quad (5.6)$$

Expanding (5.5), using $|f(\sigma)| = 1$ and (5.6), yields

$$2a(\sigma) - f(\sigma)c(\sigma) = 0 \quad (5.7)$$

almost everywhere on the arc. Every nonempty subarc has positive arclength, so the set on which (5.7) holds is dense; continuity gives (5.7) on the whole arc. Near any point of the arc, choose a biholomorphic coordinate that maps the arc to a real interval. Convexity places Ω in one half-disk, and the spectrum is a positive distance away, so $cf - 2a$ is analytic in that half-disk and continuous to its diameter. Extend it by zero to the opposite half-disk. The extension is continuous, and Morera's theorem, applied after subdividing each triangle along the diameter, makes it holomorphic. The identity theorem first gives $cf - 2a = 0$ in an interior collar. The punctured domain $\Omega \setminus \operatorname{spec}(A)$ is connected; applying the identity theorem there proves $cf - 2a = 0$ throughout it, hence as a meromorphic identity on Ω . Since $a(z) = z^{-1} + O(z^{-2})$ at infinity, a is not the zero rational function; the identity therefore also shows that c is not identically zero. This proves (5.2).

The adjugate formulas are

$$a = \frac{D_A}{\det(zI - A)}, \quad c = \frac{C_A}{\det(zI - A)}.$$

The polynomial D_A has degree exactly $N - 1$, with leading coefficient 1. Since $x^*y = 0$, the degree of C_A is at most $N - 2$. Let G be a greatest common divisor of C_A and D_A , and choose

$$U = \frac{2D_A}{G}, \quad V = \frac{C_A}{G}.$$

Then U, V are coprime, $f = U/V$, and $UC_A = 2D_A V$. Canceling G preserves the strict degree difference, proving (5.3). $\square$

6. The full lemniscate is the boundary

Lemma 9 (Zeros of the transfer function). *Let $f = U/V$ be the reduced rational function from Proposition 3. Then every finite zero of U lies in Ω , every zero of V lies outside K , and $f(\infty) = \infty$.*

Proof. For $z \notin K$, put $R = (zI - A)^{-1}$. Then $a(z) \neq 0$. Indeed, if $a(z) = x^*Rx = 0$ and $v = Rx$, then $v \neq 0$ and $(zI - A)v = x$. Since $x^*v = a(z) = 0$, conjugation gives $v^*x = 0$, and hence

$$v^*(zI - A)v = v^*x = 0, \quad \text{hence} \quad \frac{v^*Av}{\|v\|^2} = z,$$

which would put z in $W(A) = K$, a contradiction.

By the construction in Proposition 3, $U = 2D_A/G$, so every zero of U is a zero of D_A . Since $\det(zI - A) \neq 0$ off K and $a = D_A/\det(zI - A)$ does not vanish there, U has no zero off K . Nor can V vanish at a boundary point: by coprimality, U/V would then be unbounded along approaches from Ω ,

where it equals the original function $f \in \mathcal{A}(K)$. Thus U/V extends continuously to ∂K , agrees there with f , and has modulus one. In particular, U has no boundary zero, so all its zeros lie in Ω . A reduced pole cannot lie in Ω , where f is holomorphic. Hence all poles lie outside K . Finally (5.3) says exactly that $f(\infty) = \infty$. $\square$

Proposition 4 (Full-level identity). *For the reduced rational extremizer $f = U/V$,*

$$\{z \in \mathbb{C} : V(z) \neq 0, |U(z)/V(z)| < 1\} = \Omega, \quad \{z \in \mathbb{C} : V(z) \neq 0, |U(z)/V(z)| = 1\} = \partial K. \quad (6.1)$$

Moreover f' does not vanish on ∂K , and ∂K is a smooth real-analytic strictly convex Jordan curve.

Proof. Put

$$\mathcal{O} := \{z \in \mathbb{C} : V(z) \neq 0, |U(z)/V(z)| < 1\}.$$

No pole belongs to $\overline{\mathcal{O}}$, because $|f| > 1$ in a punctured neighborhood of each pole. If $z \in \partial\mathcal{O}$, continuity and approximation from $\mathcal{O}$ give $|f(z)| \leq 1$; strict inequality would put a neighborhood of z in $\mathcal{O}$. Therefore

$$\partial\mathcal{O} \subset \{z \in \mathbb{C} : V(z) \neq 0, |U(z)/V(z)| = 1\}. \quad (6.2)$$

Now $\Omega \subset \mathcal{O}$, while ∂K is disjoint from $\mathcal{O}$. Hence the connected set Ω is both open and closed relative to $\mathcal{O}$, and is one of its components.

Every component $\mathcal{O}_0$ is bounded because $f(\infty) = \infty$, and it contains a zero of f . Otherwise its compact closure would avoid all zeros of f : any zero in the closure lies in $\mathcal{O}$ and hence in $\mathcal{O}_0$. Thus $1/f$ would be holomorphic near $\overline{\mathcal{O}_0}$, have modulus one on the boundary by (6.2), and have modulus greater than one inside, contrary to the maximum-modulus principle. Lemma 9 puts every zero in Ω , so $\mathcal{O}$ has no other component. This proves the first identity in (6.1).

Conversely, if $V(z_0) \neq 0$ and $|f(z_0)| = 1$, the open mapping theorem says that every neighborhood of z_0 contains points where $|f| < 1$. Thus $z_0 \in \partial\mathcal{O}$, and (6.2) proves the second identity in (6.1).

Suppose that $f'(z_0) = 0$ at a level point. Then, in a neighborhood of z_0 ,

$$f(z) - f(z_0) = (z - z_0)^k q(z), \quad k \geq 2, \quad q(z_0) \neq 0.$$

Choose $\alpha \in \mathbb{C}$ so that $\operatorname{Re}(\overline{f(z_0)} \alpha^k q(z_0)) < 0$, and let $\zeta = e^{2\pi i/k}$. For all sufficiently small $t > 0$, the points $z_j = z_0 + t\alpha\zeta^j$, $0 \leq j < k$, are not poles and satisfy

$$|f(z_j)|^2 = 1 + 2t^k \operatorname{Re}(\overline{f(z_0)} \alpha^k q(z_0)) + O(t^{k+1}) < 1.$$

The first identity in (6.1) puts every z_j in Ω . But

$$\frac{1}{k} \sum_{j=0}^{k-1} z_j = z_0$$

because $\sum_{j=0}^{k-1} \zeta^j = 0$. Convexity of Ω would then put the boundary point z_0 in Ω , a contradiction. Hence $f' \neq 0$ on the level, and the implicit-function theorem makes the boundary real analytic and smooth.

Finally suppose that the boundary contained a nontrivial segment of a real line $z = z_0 + t\tilde{\zeta}$, $t \in \mathbb{R}$, $\tilde{\zeta} \neq 0$. The expression

$$|U(z_0 + t\tilde{\zeta})|^2 - |V(z_0 + t\tilde{\zeta})|^2$$

is a polynomial in the real variable t . Its vanishing on an interval would make it vanish for every real t . If $V(z_0 + t\tilde{\zeta}) = 0$ for some real t , the same identity would force $U(z_0 + t\tilde{\zeta}) = 0$, contrary to coprimality. The full-level identity would therefore put the entire unbounded line in ∂K , contradicting

compactness. Thus the boundary has no nontrivial segment, which for a compact convex body is exactly strict convexity. $\square$

7. The algebraic tangent argument

The Hermitian determinant below is the line-coordinate polynomial introduced by Kippenhahn [6].

Proposition 5 (Circle criterion). *Let K be a compact convex body whose boundary is the full level set*

$$\partial K = \{z \in \mathbb{C} : V(z) \neq 0, |U(z)/V(z)| = 1\}, \quad (7.1)$$

where U, V are coprime polynomials, $\deg U > \deg V$, K is strictly convex, and $(U/V)'$ does not vanish on ∂K . Writing $z = X + iY$, suppose there are Hermitian matrices H, J such that, in the real dual coordinates $[s : u : v]$ associated with the line $s + uX + vY = 0$, every tangent line to an open boundary arc belongs to the projective curve

$$\det(sI + uH + vJ) = 0. \quad (7.2)$$

Then ∂K is a circle.

Proof. Put $f = U/V$. Write $z = X + iY$, $w = X - iY$, and $S^\#(w) = \overline{S(\overline{w})}$ for a polynomial S . For a homogenization $S_h(w, Z)$, the notation $S_h^\#$ means coefficientwise conjugation, with w, Z left unchanged. We use primal coordinates $[X : Y : Z]$; a line $sZ + uX + vY = 0$ has dual coordinates $[s : u : v]$. Thus, in vector incidence formulas below, put $\mathbf{z} = (Z, X, Y)^\top$ and $\ell = (s, u, v)^\top$. Let $d = \deg U$, $e = \deg V < d$, and homogenize the real lemniscate polynomial to degree $2d$:

$$\mathcal{F}(X, Y, Z) = U_h(X + iY, Z)U_h^\#(X - iY, Z) - Z^{2(d-e)}V_h(X + iY, Z)V_h^\#(X - iY, Z). \quad (7.3)$$

Here U_h, V_h are the usual homogenizations to their own degrees. At $Z = 1$, use the Wirtinger derivative $\partial_z = \frac{1}{2}(\partial_X - i\partial_Y)$. Then

$$\mathcal{F} = |V|^2(|f|^2 - 1), \quad \partial_z \mathcal{F} = |V|^2 f' \bar{f} \quad \text{on } |f| = 1.$$

Here $V \neq 0$ on the level and $f' \neq 0$ there by hypothesis, so the boundary lies in the regular locus of $\mathcal{F} = 0$. It therefore lies in the zero set of a single complex irreducible factor of $\mathcal{F}$; denote the corresponding projective curve by $\mathcal{C}$. Locally only one factor can vanish at a regular point, and connectedness selects the same factor along the boundary. Because this factor contains a real arc, it is fixed by conjugation and can be chosen with real coefficients.

Every real affine point of $\mathcal{C}$ is a real zero of (7.3), hence is on the full level (7.1); coprimality prevents U, V from vanishing there simultaneously. Thus

$$\mathcal{C}(\mathbb{R}) \cap \{Z \neq 0\} = \{[X : Y : 1] : X + iY \in \partial K\}$$

after identifying the affine chart with $Z = 1$. There are no additional real points at infinity, because at $Z = 0$, where $\text{lc}(U)$ denotes the leading coefficient of U ,

$$\mathcal{F}(X, Y, 0) = |\text{lc}(U)|^2(X^2 + Y^2)^d, \quad (7.4)$$

which has no real projective zero. Therefore the real projective locus of $\mathcal{C}$ is exactly the one oval identified with ∂K .

Let $\mathcal{C}^*$ be the projective dual curve, the Zariski closure of the tangent lines at smooth points of $\mathcal{C}$, and let $\mathcal{Q}(s, u, v)$ be its irreducible homogeneous equation, chosen with real coefficients. Tangency on an open real arc and (7.2) imply

$$\mathcal{Q}(s, u, v) \text{ divides } \det(sI + uH + vJ). \quad (7.5)$$

Indeed, the corresponding tangent arc is Zariski dense in the irreducible curve $\mathcal{C}^*$. The determinant therefore vanishes on all of $\mathcal{C}^*$. Its homogeneous prime ideal is the principal ideal $(\mathcal{Q})$, which proves the divisibility.

Set $\mathbf{e} = [1 : 0 : 0]$. The determinant in (7.5) equals 1 at $\mathbf{e}$, so $\mathcal{Q}(\mathbf{e}) \neq 0$. For real u, v , all roots in s of the determinant are real, because they are the negatives of the eigenvalues of the Hermitian matrix $uH + vJ$. Hence every root of $\mathcal{Q}(s, u, v)$ is real. In other words, $\mathcal{Q}$ is hyperbolic with respect to $\mathbf{e}$.

Let $m = \deg \mathcal{Q}$. The coefficient of s^m in $\mathcal{Q}(s, u, v)$ is $\mathcal{Q}(1, 0, 0) \neq 0$. Choose a real projective direction $[u : v]$ outside the finite exceptional set described next. On the normalization of $\mathcal{C}$, the Gauss map to $\mathcal{C}^*$ is generically one-to-one by projective biduality in characteristic zero. Its ramification locus, the singular locus of $\mathcal{C}^*$, and the tangent images of the finitely many singular points of $\mathcal{C}$ are finite. Exclude their projections to $[u : v]$, as well as the discriminant directions of $\mathcal{Q}(s, u, v)$. The discriminant is not identically zero because the generic projection of an irreducible curve is separable in characteristic zero. The specialization then has m distinct roots, and hyperbolicity makes the corresponding dual points $\ell_j = [s_j : u : v]$ real and smooth.

Biduality gives each ℓ_j a unique contact point on the smooth locus of $\mathcal{C}$. Writing subscripts for partial derivatives, Euler's identity for the homogeneous polynomial $\mathcal{Q}$ shows that the tangent to $\mathcal{C}^*$ at ℓ_j corresponds, in the standard primal coordinate order, to

$$[X : Y : Z] = [\mathcal{Q}_u(\ell_j) : \mathcal{Q}_v(\ell_j) : \mathcal{Q}_s(\ell_j)],$$

so the contact point is real. It therefore belongs to the oval identified with ∂K .

A smooth strictly convex oval has exactly two tangent lines with a prescribed unoriented normal $[u : v]$: the support line at the maximum and the one at the minimum of $uX + vY$. The preceding count shows that there are at most two roots; conversely these two support lines belong to $\mathcal{C}^*$, so both are roots. Consequently

$$m = 2. \quad (7.6)$$

Thus $\mathcal{C}^*$ is an irreducible conic and is nonsingular. Writing its equation as $\ell^T \Sigma \ell = 0$, with an invertible symmetric 3×3 matrix Σ , its tangent at ℓ corresponds to the primal point $\Sigma \ell$. The envelope of those tangents is $\mathbf{z}^T \Sigma^{-1} \mathbf{z} = 0$, also a conic. Since it contains a Zariski-dense arc of $\mathcal{C}$, it is $\mathcal{C}$.

Finally, because the defining polynomial of $\mathcal{C}$ divides $\mathcal{F}$, every point of this conic at infinity must be one of

$$[1 : i : 0], \quad [1 : -i : 0].$$

Bézout's theorem applied to the line $Z = 0$ gives two intersection points counted with multiplicity. Complex conjugation exchanges the two displayed, distinct circular points, so each occurs once. The real homogeneous equation of the conic therefore has the form

$$q_0(X^2 + Y^2) + q_1 XZ + q_2 YZ + q_3 Z^2 = 0, \quad q_0 \neq 0.$$

Completing squares shows that its nonempty compact real oval is a circle. $\square$

8. Proof of the main theorem

Proof of Theorem 1. We use induction on N . For $N = 1$, and more generally whenever the numerical range has empty interior, Lemma 1 gives $\psi(A) = 1$, so the hypothesis never occurs.

Assume the theorem holds for all sizes smaller than N , and let $A \in M_N(\mathbb{C})$ satisfy $\psi(A) = 2$. Put $K = W(A)$. It has nonempty interior. By Lemma 6, either K is already a nondegenerate disk or $\text{spec}(A) \subset \Omega = \text{int } K$. Consider the second case.

Choose $f \in \mathcal{A}(K)$ as in Lemma 7, and let x, y be the equality vectors from Proposition 2. Since an interior-spectrum numerical range cannot be a polygon, Lemma 8 provides a curved analytic boundary arc. On that arc, Proposition 3 yields

$$f(z) = \frac{U(z)}{V(z)}, \quad \deg U > \deg V,$$

after cancellation of the meromorphic transfer identity (5.2).

Lemma 9 puts every zero of U in Ω , every pole outside K , and gives $f(\infty) = \infty$. Proposition 4 then gives

$$\{z \in \mathbb{C} : V(z) \neq 0, |U(z)/V(z)| < 1\} = \Omega, \quad \{z \in \mathbb{C} : V(z) \neq 0, |U(z)/V(z)| = 1\} = \partial K,$$

and shows that this boundary is a smooth strictly convex rational lemniscate.

Put $H_A = \text{Re } A$ and $J_A = \text{Im } A$. Writing $z = X + iY$, every supporting line $s + uX + vY = 0$, with $s, u, v \in \mathbb{R}$, can be oriented so that $sI + uH_A + vJ_A \succeq 0$. A unit vector attaining the point of support has zero quadratic form for this positive semidefinite matrix and therefore lies in its kernel. Hence

$$\det(sI + uH_A + vJ_A) = 0.$$

All hypotheses of Proposition 5 now hold. It follows that ∂K is a circle. Compactness and convexity make K the filled closed disk bounded by that circle.

The disk has positive radius because K has nonempty planar interior. Writing its center and radius as γ and ρ gives

$$W(A) = \{z : |z - \gamma| \leq \rho\}, \quad \rho > 0,$$

which completes the induction and the proof. $\square$

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